

## EDUCATION

## Columbia University

NEW YORK

B.A candidate in Computer Science and Mathematics. GPA: 3.93/4.00

AUG 2022 - PRESENT

Selected Coursework: COMS 6998 Theory of Large Language Models, COMS 4774 Unsupervised Learning, COMS 4771 Machine Learning, COMS4731 Computer Vision, CSOR4231W Analysis of Algorithms

## Raffles Institution

SINGAPORE

Singapore-Cambridge GCE A-Levels: 6 Distinctions

JAN 2014 - DEC 2019

- Coursework: Physics(H2 and H3), Chemistry, Math, Economics, General Paper
- Enrolled in Raffles Academy, a highly competitive and selective advanced STEM program

Science Society - Vice Chairman

- Oversaw the planning and organization of events such as the Science Olympiad held by the school, Science Advocacy week, speaker events etc.. Prepared weekly club sessions for club members

## SELECTED AWARDS AND ACHIEVEMENTS

## International Olympiad on Astronomy and Astrophysics: Bronze Medal

AUG 2019

- Achieved a gold medal and an individual 5th ranking in the National Astronomy Olympiad. Represented Singapore at the International Olympiad on Astronomy and Astrophysics and achieved a bronze medal

## Singapore Space Challenge: Overall Champions

JUN 2019

- Designed and modeled a CubeSatellite for use in the detection of wildfires. Awarded the grand prize and a monetary prize of SGD10,000

## Singapore Physics Olympiad: Silver Medal

NOV 2018

- Achieved a silver medal in the National Physics Olympiad

## MIT Pokerbots 2024: Rank 11th/96 Teams

JAN 2024

- Built a Python bot that played a variant of heads-up poker against opponent bots; overall win rate of 73%

## QUANTIACS Q18 Algorithmic Trading Contest: Rank 9th

MAR 2024

- Coded, back-tested and live-tested several algorithms for trading stocks using Python

## PROFESSIONAL AND RESEARCH EXPERIENCE

## Complex Resilient Intelligent Systems Laboratory (CRIS Lab): Research Assistant

NEW YORK

- Conducted research on the mechanisms behind self-organization and the emergence of collective intelligence in deep neural networks

AUG 2023 - DEC 2024

## Luce Maintenance Group Singapore: Software Engineering Intern

SINGAPORE

- Developed and deployed end-to-end features, including frontend UI components using React and backend APIs with Node.js, to enhance user experience
- Worked on optimizing codebase to reduce computational costs and execution speeds for Ruby on Rails web service app. Enhanced database and querying efficiency, resulting in a 30% reduction in server loads

JUN 2023 - AUG 2023

## AlQuraishi Lab at Columbia University: Research Assistant

NEW YORK

- Investigated the reimplementation of AlphaFold2 (a machine learning framework developed for predicting protein structure) to create a memory efficient and trainable method for predicting intermolecular interactions

MAR 2023 - AUG 2023

## Victoris: Co-Founder

SINGAPORE

- Co-Founder of an education group providing academic enrichment programmes to over 150 students
- Organized talks and information sessions for disadvantaged children, including topics such as College Admissions and Information Access

JUN 2021 - AUG 2022

## Republic of Singapore Air Force: Immediate Response Force (mandatory military service)

SINGAPORE

- Performed security operations at Paya Lebar Airbase as a member of the Immediate Response Force under the Perimeter Protection Flight

JAN 2020 - NOV 2021

## Defence Science Organization: Research Intern

SINGAPORE

- Designed and created numerical models of reflectarray antennas using COMSOL Multiphysics, Python, MATLAB and AutoCad that achieved perfect agreement with experimental data
- Worked with PCB fabrication and 3D printing technology to construct physical models of the antennas for measurement and testing in the Electromagnetics Laboratory

MAY 2018 - JAN 2019

## SKILLS

**Proficient:** Python, Mathematica. **Intermediate:** C, C++, Java, Javascript, HTML, Ruby, MATLAB, COMSOL, Microsoft Excel, AutoCAD, PostgreSQL

**Languages:** Native proficiency in English and Chinese. Limited working proficiency in Japanese.